Ex 7: Model Planning and Building – Clustering

Question

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence and Data Science
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Ex7: Model Planning and Building - Clustering

Learning Objective:
The objective of this experiment is to enable students to understand and implement the K-Means clustering algorithm for customer segmentation.

Problem Statement:
A retail company wants to group its customers based on their Annual Income and Spending Score to design personalized marketing campaigns. By identifying customers with similar purchasing behavior, the company can improve customer satisfaction, optimize promotional offers, and increase overall sales.

Your task is to analyze the customer dataset, apply the K-Means Clustering algorithm to identify customer segments, visualize the clusters, and recommend suitable marketing strategies for each segment.

Tasks to Perform:

1. Use Mall Customer Segmentation Dataset from Kaggle and load it into a Pandas DataFrame.
2. Explore the dataset by: Displaying the first and last five records, Checking the dataset shape and data types, Identifying missing values and duplicate records, Displaying summary statistics.
3. Select the important features for clustering: Annual Income (k$), Spending Score (1–100).
4. Standardize the selected features using StandardScaler for clustering.
5. Visualize the selected features using: Scatter Plot, Pair Plot.

Model Planning and Building

6. Determine the optimal number of clusters using the Elbow Method. Plot the Elbow Curve and identify the optimal value of K.
7. Train the K-Means Clustering model using the optimal number of clusters. Assign cluster labels to each customer, add as a new column in the dataset and save as a file.

Cluster Visualization

8. Visualize the customer clusters using a scatter plot: X-axis: Annual Income, Y-axis: Spending Score, Different colors for each cluster, Highlight cluster centroids, Display the coordinates of the cluster centroids.

Cluster Analysis

9. Calculate the number of customers in each cluster.
10. Compute the average Annual Income and Spending Score for each cluster.

Aim

To segment mall customers by Annual Income and Spending Score using the K-Means clustering algorithm and recommend marketing strategies for each segment.

Procedure

1. Load the Mall Customer Segmentation dataset from Kaggle.
2. Explore the data: first and last five rows, shape, data types, missing values, duplicates and summary statistics.
3. Select Annual Income (k$) and Spending Score (1–100) as clustering features and standardise them with StandardScaler.
4. Visualise the features with a scatter plot and a pair plot.
5. Use the Elbow Method (WCSS vs K) to choose the optimal K.
6. Train K-Means with the optimal K, add the cluster labels as a new column and save the file.
7. Plot the clusters with centroids highlighted and display the centroid coordinates.
8. Count customers per cluster and compute the mean income and spending score of each cluster.

Code

```
#Mohammed Fayyad(241801165)
import pandas as pd, numpy as np, matplotlib.pyplot as plt, seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

df = pd.read_csv("Mall_Customers.csv")
print(df.head()); print(df.tail()); print(df.shape); print(df.dtypes)
print(df.isnull().sum()); print("Duplicates:", df.duplicated().sum());
print(df.describe())

features = ["Annual Income (k$)", "Spending Score (1-100)"]
X = df[features]
X_scaled = StandardScaler().fit_transform(X)

plt.scatter(X[features[0]], X[features[1]]); plt.xlabel(features[0]);
plt.ylabel(features[1]); plt.show()
sns.pairplot(df[features]); plt.show()

# Elbow method
wcss = []
for k in range(1, 11):
    wcss.append(KMeans(n_clusters=k, init="k-means++", n_init=10,
random_state=42).fit(X_scaled).inertia_)
plt.plot(range(1, 11), wcss, marker="o"); plt.xlabel("K"); plt.ylabel("WCSS");
plt.title("Elbow Curve"); plt.show()
```

```
k = 5   # optimal K read from the elbow curve
km = KMeans(n_clusters=k, init="k-means++", n_init=10, random_state=42)
df["Cluster"] = km.fit_predict(X_scaled)
df.to_csv("Mall_Customers_Clustered.csv", index=False)

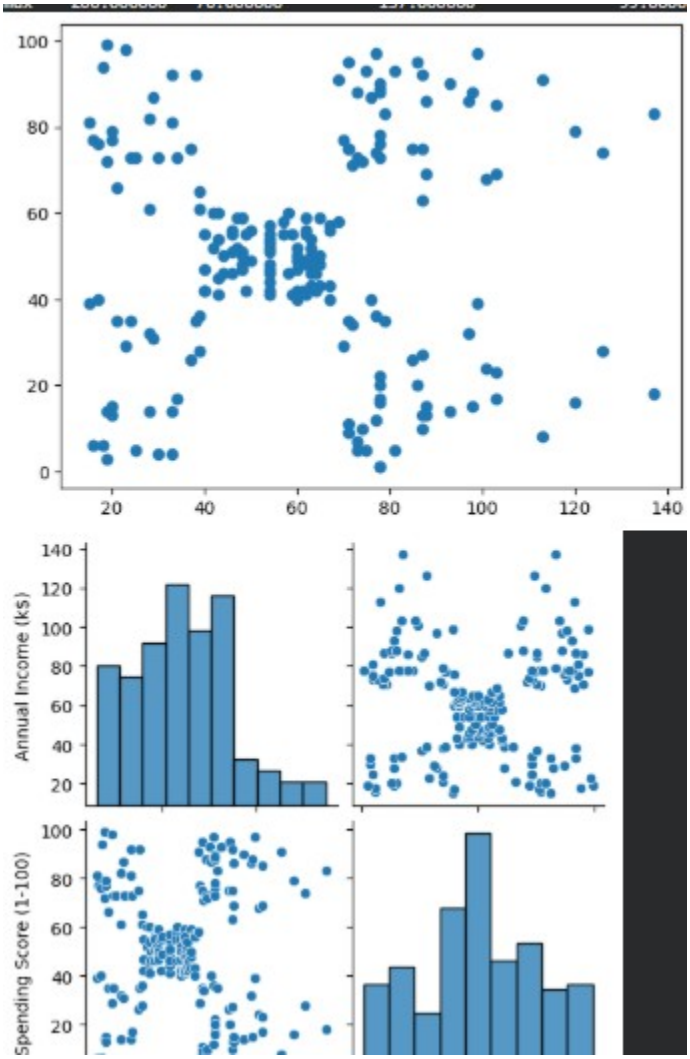
# Centroids back in original units
centroids = StandardScaler().fit(X).inverse_transform(km.cluster_centers_)
print("Centroids (Income, Score):\n", centroids)

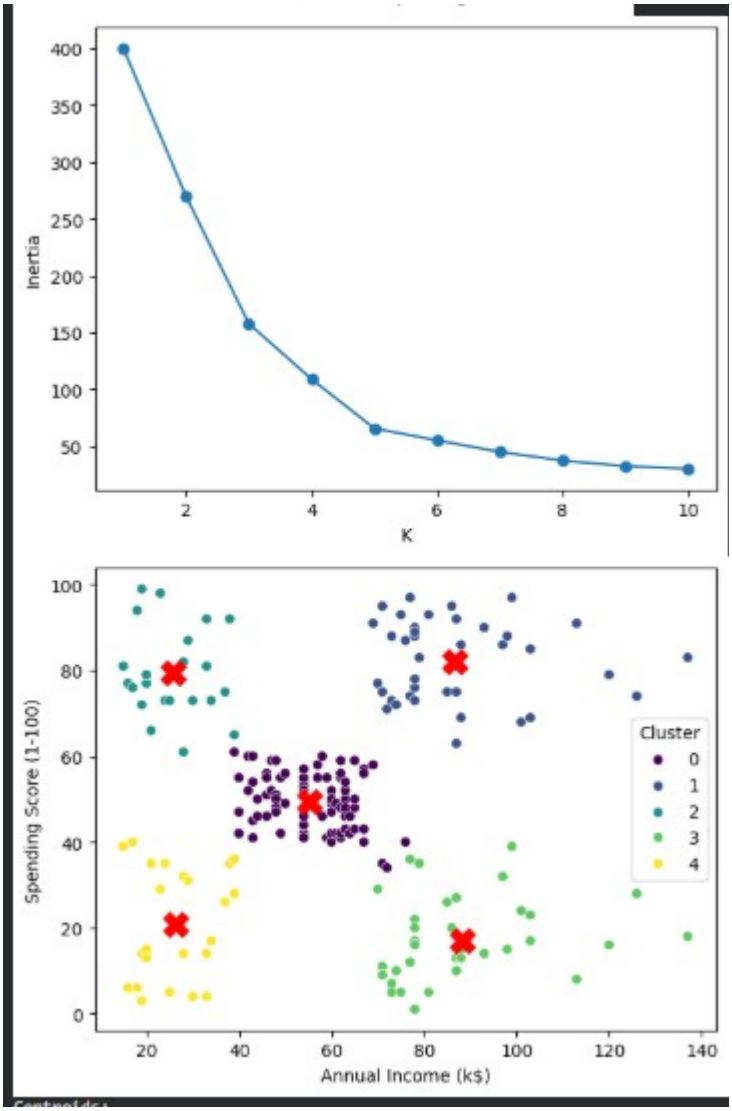
plt.figure(figsize=(8, 6))
for c in range(k):
    sub = df[df.Cluster == c]
    plt.scatter(sub[features[0]], sub[features[1]], label=f"Cluster {c}")
plt.scatter(centroids[:, 0], centroids[:, 1], s=300, c="black", marker="X",
label="Centroids")
plt.xlabel("Annual Income (k$)"); plt.ylabel("Spending Score"); plt.legend();
plt.show()

print(df["Cluster"].value_counts().sort_index())
print(df.groupby("Cluster")[features].mean())
```

Result

The Elbow curve gave the optimal number of clusters, K-Means grouped customers into distinct segments, the clusters and centroids were plotted, and the customers per cluster along with the mean Annual Income and Spending Score of each cluster were computed. Marketing strategies can be designed for each segment accordingly.


Annual Income (k$)
Spending Score (1-100)


Inertia
K
Spending Score (1-100)
Annual Income (k$)
Cluster
0
1
2
3
4